MSG No. 211.58 $I4'32'$ [Type III, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 4'32'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 6b, site symmetry: 4'2.2'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,13,14,21,22]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,8,10,12,17,18,19,20]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,7,9,11,15,16,23,24]
4	$\left[\frac{1}{2}, 0, 0\right]$	[25,26,27,28,37,38,45,46]
5	$\left[0, \frac{1}{2}, 0\right]$	[29,32,34,36,41,42,43,44]
6	$\left[0, 0, \frac{1}{2}\right]$	[30,31,33,35,39,40,47,48]

Table 3: Wyckoff site: 8c, site symmetry: .32'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,6,44,46,48]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,40,41,45]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,7,12,37,42,47]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,8,9,38,39,43]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[13,18,23,27,31,36]
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[14,15,19,28,32,33]
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[16,17,21,26,34,35]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[20,22,24,25,29,30]

Table 4: Wyckoff site: 12d, site symmetry: 2.2'2'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1,2,45,46]
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[3,4,37,38]
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[5,12,42,44]
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[6,9,39,48]
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,40,47]
6	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[8,10,41,43]
7	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[13,14,27,28]
8	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[15,24,30,33]
9	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[16,23,31,35]
10	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[17,19,32,34]
11	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[18,20,29,36]
12	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[21,22,25,26]

Table 5: Wyckoff site: `12e`, site symmetry: `4'..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,13,14]`
2	$[-x,\ 0,\ 0]$	`[3,4,21,22]`
3	$[0,\ x,\ 0]$	`[5,12,17,19]`
4	$[0,\ 0,\ x]$	`[6,9,16,23]`
5	$[0,\ 0,\ -x]$	`[7,11,15,24]`
6	$[0,\ -x,\ 0]$	`[8,10,18,20]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[25,26,37,38]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[27,28,45,46]`
9	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[29,36,41,43]`
10	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[30,33,40,47]`
11	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[31,35,39,48]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[32,34,42,44]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,6]`
2	$[x,\ -x,\ -x]$	`[2,10,11]`
3	$[-x,\ x,\ -x]$	`[3,7,12]`
4	$[-x,\ -x,\ x]$	`[4,8,9]`
5	$[x,\ -x,\ x]$	`[13,18,23]`
6	$[x,\ x,\ -x]$	`[14,15,19]`
7	$[-x,\ x,\ x]$	`[16,17,21]`
8	$[-x,\ -x,\ -x]$	`[20,22,24]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[25,29,30]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[26,34,35]`
11	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[27,31,36]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[28,32,33]`
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[37,42,47]`
14	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[38,39,43]`
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[40,41,45]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[44,46,48]`

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2]`
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,4]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,12]`
4	$\left[\frac{1}{2},\ 0,\ x\right]$	`[6,9]`
5	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[7,11]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[8,10]`
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[13,14]`
8	$\left[0,\ \frac{1}{2},\ -x\right]$	`[15,24]`

continued ...

Table 7

No.	position	mapping
9	$\left[0,\ \frac{1}{2},\ x\right]$	[16,23]
10	$\left[\frac{1}{2},\ x,\ 0\right]$	[17,19]
11	$\left[\frac{1}{2},\ -x,\ 0\right]$	[18,20]
12	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[21,22]
13	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[25,26]
14	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[27,28]
15	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[29,36]
16	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[30,33]
17	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[31,35]
18	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[32,34]
19	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[37,38]
20	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[39,48]
21	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[40,47]
22	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[41,43]
23	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[42,44]
24	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[45,46]

Table 8: Wyckoff site: 24h, site symmetry: ..2'

No.	position	mapping
1	$[0,\ y,\ y]$	[1,21]
2	$[0,\ -y,\ -y]$	[2,22]
3	$[0,\ y,\ -y]$	[3,14]
4	$[0,\ -y,\ y]$	[4,13]
5	$[y,\ 0,\ y]$	[5,18]
6	$[y,\ y,\ 0]$	[6,15]
7	$[-y,\ y,\ 0]$	[7,16]
8	$[-y,\ 0,\ y]$	[8,17]
9	$[-y,\ -y,\ 0]$	[9,24]
10	$[y,\ 0,\ -y]$	[10,19]
11	$[y,\ -y,\ 0]$	[11,23]
12	$[-y,\ 0,\ -y]$	[12,20]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[25,45]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[26,46]
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[27,38]
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[28,37]
17	$\left[y+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[29,42]
18	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[30,39]
19	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[31,40]
20	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[32,41]
21	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[33,48]
22	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[34,43]
23	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[35,47]
24	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[36,44]

Table 9: Wyckoff site: `24i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	`[1,46]`
2	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	`[2,45]`
3	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	`[3,37]`
4	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	`[4,38]`
5	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	`[5,44]`
6	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[6,48]`
7	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[7,47]`
8	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	`[8,43]`
9	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,39]`
10	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	`[10,41]`
11	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[11,40]`
12	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	`[12,42]`
13	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$	`[13,27]`
14	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$	`[14,28]`
15	$\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$	`[15,33]`
16	$\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[16,35]`
17	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	`[17,34]`
18	$\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	`[18,36]`
19	$\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	`[19,32]`
20	$\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	`[20,29]`
21	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$	`[21,26]`
22	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$	`[22,25]`
23	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$	`[23,31]`
24	$\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[24,30]`

Table 10: Wyckoff site: `48j`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[y,\ z,\ x]$	`[6]`
7	$[-y,\ z,\ -x]$	`[7]`
8	$[-z,\ -x,\ y]$	`[8]`
9	$[-y,\ -z,\ x]$	`[9]`
10	$[z,\ -x,\ -y]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-z,\ x,\ -y]$	`[12]`
13	$[x,\ -z,\ y]$	`[13]`
14	$[x,\ z,\ -y]$	`[14]`
15	$[z,\ y,\ -x]$	`[15]`
16	$[-z,\ y,\ x]$	`[16]`
17	$[-y,\ x,\ z]$	`[17]`

continued ...

Table 10

No.	position	mapping
18	$[y,\,-x,\,z]$	[18]
19	$[y,\,x,\,-z]$	[19]
20	$[-y,\,-x,\,-z]$	[20]
21	$[-x,\,z,\,y]$	[21]
22	$[-x,\,-z,\,-y]$	[22]
23	$[z,\,-y,\,x]$	[23]
24	$[-z,\,-y,\,-x]$	[24]
25	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,\frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,\frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\,x+\frac{1}{2},\,y+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\,z+\frac{1}{2},\,x+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-y,\,z+\frac{1}{2},\,\frac{1}{2}-x\right]$	[31]
32	$\left[\frac{1}{2}-z,\,\frac{1}{2}-x,\,y+\frac{1}{2}\right]$	[32]
33	$\left[\frac{1}{2}-y,\,\frac{1}{2}-z,\,x+\frac{1}{2}\right]$	[33]
34	$\left[z+\frac{1}{2},\,\frac{1}{2}-x,\,\frac{1}{2}-y\right]$	[34]
35	$\left[y+\frac{1}{2},\,\frac{1}{2}-z,\,\frac{1}{2}-x\right]$	[35]
36	$\left[\frac{1}{2}-z,\,x+\frac{1}{2},\,\frac{1}{2}-y\right]$	[36]
37	$\left[x+\frac{1}{2},\,\frac{1}{2}-z,\,y+\frac{1}{2}\right]$	[37]
38	$\left[x+\frac{1}{2},\,z+\frac{1}{2},\,\frac{1}{2}-y\right]$	[38]
39	$\left[z+\frac{1}{2},\,y+\frac{1}{2},\,\frac{1}{2}-x\right]$	[39]
40	$\left[\frac{1}{2}-z,\,y+\frac{1}{2},\,x+\frac{1}{2}\right]$	[40]
41	$\left[\frac{1}{2}-y,\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[41]
42	$\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[42]
43	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}-z\right]$	[43]
44	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,\frac{1}{2}-z\right]$	[44]
45	$\left[\frac{1}{2}-x,\,z+\frac{1}{2},\,y+\frac{1}{2}\right]$	[45]
46	$\left[\frac{1}{2}-x,\,\frac{1}{2}-z,\,\frac{1}{2}-y\right]$	[46]
47	$\left[z+\frac{1}{2},\,\frac{1}{2}-y,\,x+\frac{1}{2}\right]$	[47]
48	$\left[\frac{1}{2}-z,\,\frac{1}{2}-y,\,\frac{1}{2}-x\right]$	[48]